Assignment 2

Study of Linux Networking Commands

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1. ifconfig

Description: ifconfig is used to display and configure network interface parameters. It allows you to view your current network settings (like IP addresses and MAC addresses) and can be used to manually enable or disable network interfaces. It is also used at boot time to set up interfaces as necessary.

Output:

```
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ ifconfig
lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 3902  bytes 426181 (426.1 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 3902  bytes 426181 (426.1 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

wlo1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.1.14  netmask 255.255.255.0  broadcast 192.168.1.255
        inet6 fe80::b7cd:6fb0:ac76:b988  prefixlen 64  scopeid 0x20<link>
        inet6 2401:4900:8fc8:a787:6656:b002:8efc:7e9b  prefixlen 64  scopeid 0x0<global>
        inet6 2401:4900:8fc8:a787:740b:8ff:c524:75c3  prefixlen 64  scopeid 0x0<global>
        ether a8:41:f4:af:ad:31  txqueuelen 1000  (Ethernet)
        RX packets 529241  bytes 630266667 (630.2 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 92324  bytes 32350153 (32.3 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

2. ip

Description: ip is used to show and manipulate routing, network devices, interfaces and tunnels.

Output:

```
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: wlo1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether a8:41:f4:af:ad:31 brd ff:ff:ff:ff:ff:ff
    altname wlp1s0
    inet 192.168.1.14/24 brd 192.168.1.255 scope global dynamic noprefixroute wlo1
       valid_lft 83697sec preferred_lft 83697sec
    inet6 2401:4900:8fc8:a787:6656:b002:8efc:7e9b/64 scope global temporary dynamic
       valid_lft 86160sec preferred_lft 83155sec
    inet6 2401:4900:8fc8:a787:740b:8ff:c524:75c3/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86160sec preferred_lft 86160sec
    inet6 fe80::b7cd:6fb0:ac76:b988/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
arnav@arnav-HP-Laptop-15s-fq5xxx:~$
```

3. traceroute

Description: traceroute is a tool which is used to track the real-time path that a packet takes from a source to a destination. It records the IP address of every router (hop) it passes through, measuring the RTT (round-trip time) at each point along the route.

Output:

```
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ traceroute coeptech.ac.in
traceroute to coeptech.ac.in (139.59.7.176), 30 hops max, 60 byte packets
 1  local.airtelfiber.com (192.168.1.1)  8.779 ms  8.689 ms  8.659 ms
 2  10.240.8.56 (10.240.8.56)  8.632 ms  8.603 ms  8.576 ms
 3  * * *
 4  dsl-del-133.25.246.61.airtelbroadband.in (61.246.25.133)  8.364 ms  8.380 ms dsl-del-129.25.24
6.61.airtelbroadband.in (61.246.25.129)  8.274 ms
 5  116.119.44.252 (116.119.44.252)  31.599 ms  28.400 ms 182.79.153.0 (182.79.153.0)  34.797 ms
 6  182.79.27.226 (182.79.27.226)  31.492 ms  26.235 ms  26.183 ms
 7  143.244.224.242 (143.244.224.242)  32.689 ms  29.412 ms *
 8  * * *
 9  * * *
10  139.59.7.176 (139.59.7.176)  32.468 ms !X  35.766 ms !X  31.904 ms !X
arnav@arnav-HP-Laptop-15s-fq5xxx:~$
```

4. tracepath

Description: tracepath is a utility used to trace the path to a destination host while discovering the MTU (Maximum Transmission Unit) along the path. Unlike traceroute, it does not require superuser (root) privileges to run because it uses random UDP ports instead of ICMP or specific raw sockets.

Output:

```
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ tracepath coeptech.ac.in
 1?: [LOCALHOST]                      pmtu 1500
 1:  local.airtelfiber.com                                 2.303ms
 1:  local.airtelfiber.com                                 2.362ms
 2:  local.airtelfiber.com                                 3.276ms pmtu 1472
 2:  10.240.8.56                                           5.856ms
 3:  no reply
 4:  dsl-del-133.25.246.61.airtelbroadband.in              7.096ms asymm  3
 5:  182.79.153.0                                         30.103ms asymm  6
 6:  182.79.27.226                                        30.613ms
 7:  143.244.224.242                                      37.705ms asymm  6
 8:  no reply
 9:  no reply
10:  139.59.7.176                                         31.447ms !H
     Resume: pmtu 1472
arnav@arnav-HP-Laptop-15s-fq5xxx:~$
```

5. ping

Description: ping is a utility used to test the reachability of a host on an IP network and measure the RTT for packets sent to a destination. It uses ICMP Echo Request and Reply messages to verify connectivity and network stability. It also reports packet loss.

Output:

```
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ ping -c 6 google.com
PING google.com (2404:6800:4009:803::200e) 56 data bytes
64 bytes from pnbomb-ab-in-x0e.1e100.net (2404:6800:4009:803::200e): icmp_seq=1 ttl=117 time=9.12 ms
64 bytes from pnbomb-ab-in-x0e.1e100.net (2404:6800:4009:803::200e): icmp_seq=2 ttl=117 time=9.06 ms
64 bytes from pnbomb-ab-in-x0e.1e100.net (2404:6800:4009:803::200e): icmp_seq=3 ttl=117 time=11.0 ms
64 bytes from pnbomb-ab-in-x0e.1e100.net (2404:6800:4009:803::200e): icmp_seq=4 ttl=117 time=9.15 ms
64 bytes from pnbomb-ab-in-x0e.1e100.net (2404:6800:4009:803::200e): icmp_seq=5 ttl=117 time=8.98 ms
64 bytes from pnbomb-ab-in-x0e.1e100.net (2404:6800:4009:803::200e): icmp_seq=6 ttl=117 time=9.79 ms

--- google.com ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5006ms
rtt min/avg/max/mdev = 8.979/9.512/10.988/0.711 ms
arnav@arnav-HP-Laptop-15s-fq5xxx:~$
```

6. netstat

Description: netstat is a command-line utility that displays network connections (both incoming and outgoing), routing tables, and a variety of network interface and network protocol statistics. It is used for finding problems in the network and to determine the amount of traffic on the network as a performance measurement.

Output:

```
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:111             0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:41941           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:2049            0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:59465           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:43735           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:54945           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:39595           0.0.0.0:*               LISTEN
tcp6       0      0 :::22                   :::*                    LISTEN
tcp6       0      0 :::111                  :::*                    LISTEN
tcp6       0      0 :::36641                :::*                    LISTEN
tcp6       0      0 ::1:631                 :::*                    LISTEN
tcp6       0      0 :::2049                 :::*                    LISTEN
tcp6       0      0 :::35557                :::*                    LISTEN
tcp6       0      0 :::46277                :::*                    LISTEN
tcp6       0      0 :::39669                :::*                    LISTEN
tcp6       0      0 :::39439                :::*                    LISTEN
udp        0      0 0.0.0.0:35820           0.0.0.0:*
udp        0      0 0.0.0.0:45800           0.0.0.0:*
udp        0      0 0.0.0.0:45826           0.0.0.0:*
udp        0      0 0.0.0.0:37808           0.0.0.0:*
udp        0      0 224.0.0.251:5353        0.0.0.0:*
udp        0      0 224.0.0.251:5353        0.0.0.0:*
udp        0      0 0.0.0.0:5353            0.0.0.0:*
udp        0      0 0.0.0.0:39140           0.0.0.0:*
udp        0      0 127.0.0.54:53           0.0.0.0:*
udp        0      0 127.0.0.53:53           0.0.0.0:*
udp        0      0 0.0.0.0:111             0.0.0.0:*
udp        0      0 0.0.0.0:41758           0.0.0.0:*
```

```
udp        0      0 127.0.0.1:849           0.0.0.0:*
udp6       0      0 :::5353                 :::*
udp6       0      0 :::39751                :::*
udp6       0      0 :::40264                :::*
udp6       0      0 :::48566                :::*
udp6       0      0 :::40387                :::*
udp6       0      0 :::48710                :::*
udp6       0      0 :::111                  :::*
udp6       0      0 fe80::b7cd:6fb0:ac7:546 :::*
udp6       0      0 :::34931                :::*
arnav@arnav-HP-Laptop-15s-fq5xxx:~$
```

7. nslookup

Description: nslookup is a program to query Internet domain name servers. It has two modes: interactive and non-interactive. Interactive mode allows the user to query name servers for information about various hosts and domains or to print a list of hosts in a domain. Non-interactive mode prints just the name and requested information for a host or domain.

Output:

```
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ nslookup google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.70.78
Name:   google.com
Address: 2404:6800:4009:803::200e

arnav@arnav-HP-Laptop-15s-fq5xxx:~$
```

8. dig

Description: dig is a flexible tool for interrogating DNS name servers. It performs DNS lookups and displays the answers that are returned from the name server(s) that were queried. Network administrators use dig to troubleshoot DNS problems because of its flexibility, ease of use and clarity of output.

Output:

```
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ dig google.com

; <<>> DiG 9.18.30-0ubuntu0.24.04.2-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 46327
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;google.com.                    IN      A

;; ANSWER SECTION:
google.com.             193     IN      A       142.250.71.110

;; Query time: 6 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Feb 02 20:17:56 IST 2026
;; MSG SIZE  rcvd: 55

arnav@arnav-HP-Laptop-15s-fq5xxx:~$
```

9. route

Description: route is a command-line utility used to view and manipulate the IP routing table in the Linux kernel. Its primary function is to set up static routes to specific hosts or networks via an interface after it has been configured with the ifconfig program.

Output:

```
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ route
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
default         local.airtelfib 0.0.0.0         UG    600    0        0 wlo1
192.168.1.0     0.0.0.0         255.255.255.0   U     600    0        0 wlo1
arnav@arnav-HP-Laptop-15s-fq5xxx:~$
```

10. host

Description: host is a simple DNS lookup utility used to translate domain names into IP addresses and perform reverse lookups. It provides a concise summary of DNS records, and is ideal for quick resolution tasks and basic DNS troubleshooting.

Output:

```
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ host google.com
google.com has address 142.250.71.110
google.com has IPv6 address 2404:6800:4009:81f::200e
google.com mail is handled by 10 smtp.google.com.
arnav@arnav-HP-Laptop-15s-fq5xxx:~$
```

11. arp

Description: arp is a utility used to view and manipulate the system's ARP cache, which stores the mapping between IPv4 addresses and their corresponding MAC (Media Access Control) addresses. It can add entries to the table, delete one or display the current content.

Output:

```
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ arp
Address                  HWtype  HWaddress           Flags Mask            Iface
192.168.1.4              ether   64:12:36:15:5d:78   C                     wlo1
local.airtelfiber.com    ether   b4:a7:c6:3d:68:80   C                     wlo1
arnav@arnav-HP-Laptop-15s-fq5xxx:~$
```

12. iwconfig

Description: iwconfig is similar to ifconfig, but it is dedicated to wireless interfaces. It is used to display and configure parameters for wireless network interfaces. While ifconfig handles general network settings, iwconfig focuses specifically on Wi-Fi-specific attributes like SSID, frequency and encryption. It may also be used to display wireless statistics.

Output:

```
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ iwconfig
lo        no wireless extensions.

wlo1      IEEE 802.11  ESSID:"Airtel_Technologicle"
          Mode:Managed  Frequency:5.24 GHz  Access Point: B4:A7:C6:3D:68:82
          Bit Rate=650 Mb/s   Tx-Power=20 dBm
          Retry short limit:7   RTS thr:off   Fragment thr:off
          Power Management:off
          Link Quality=64/70  Signal level=-46 dBm
          Rx invalid nwid:0  Rx invalid crypt:0  Rx invalid frag:0
          Tx excessive retries:1  Invalid misc:6   Missed beacon:0

arnav@arnav-HP-Laptop-15s-fq5xxx:~$ _
```

13. curl

Description: curl is a command-line tool used to transfer data to or from a server using one of the many supported protocols (such as HTTP, HTTPS, FTP, or SMTP). The command is designed to work without user interaction.

Output:

```
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ curl https://example.com
<!doctype html><html lang="en"><head><title>Example Domain</title><meta name="vi
ewport" content="width=device-width, initial-scale=1"><style>body{background:#ee
e;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-size:1.5e
m}div{opacity:0.8}a:link,a:visited{color:#348}</style><body><div><h1>Example Dom
ain</h1><p>This domain is for use in documentation examples without needing perm
ission. Avoid use in operations.<p><a href="https://iana.org/domains/example">Le
arn more</a></div></body></html>
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ 
```

14. wget

Description: wget is a free utility for non-interactive download of files from the Web. It supports HTTP, HTTPS and FTP protocols, as well as retrieval through HTTP proxies. It is optimized for downloading large files and entire directories recursively, with the ability to resume interrupted downloads and operate in the background.

Output:

```
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ wget https://example.com
--2026-02-02 20:25:00--  https://example.com/
Resolving example.com (example.com)... 2606:4700:83b2:8c7c:3342:6da:ccc4:a209, 104.18.27.120, 104.18.26.120
Connecting to example.com (example.com)|2606:4700:83b2:8c7c:3342:6da:ccc4:a209|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: ‘index.html’

index.html                [ <=>                   ]     513  --.-KB/s    in 0s

2026-02-02 20:25:00 (20.2 MB/s) - ‘index.html’ saved [513]

arnav@arnav-HP-Laptop-15s-fq5xxx:~$
```

15. telnet

Description: The telnet command is used for interactive communication with another host using the TELNET protocol. It is primarily used as a diagnostic tool to test connectivity to specific ports on a remote host.

Output:

```
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ hostname -I
192.168.1.14 2401:4900:8fc8:a787:7f46:2c38:147a:3052 2401:4900:8fc8:a787:740b:8ff:c524:75c3
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ telnet 192.168.1.17
Trying 192.168.1.17...
Connected to 192.168.1.17.
Escape character is '^]'.
Ubuntu 20.04.6 LTS
arnav-desktop-1 login: arnav
Password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-139-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Infrastructure is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

169 additional security updates can be applied with ESM Infra.
Learn more about enabling ESM Infra service for Ubuntu 20.04 at
https://ubuntu.com/20-04

New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Your Hardware Enablement Stack (HWE) is supported until April 2025.
Last login: Mon Feb  2 23:23:30 IST 2026 from 192.168.1.14 on pts/0
arnav@arnav-desktop-1:~$ hostname -I
192.168.1.17 2401:4900:8fc8:a787:5816:3c44:26d3:b1ff 2401:4900:8fc8:a787:123d:a9cc:e706:2941
arnav@arnav-desktop-1:~$ logout
Connection closed by foreign host.
arnav@arnav-HP-Laptop-15s-fq5xxx:~$
```

16. whois

Description: whois is a utility used to query databases that store registration and ownership information for domain names and IP addresses. It is primarily used to identify domain owners, check expiration dates and determine which organization an IP block is assigned to.

Output:

```
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ whois example.com
   Domain Name: EXAMPLE.COM
   Registry Domain ID: 2336799_DOMAIN_COM-VRSN
   Registrar WHOIS Server: whois.iana.org
   Registrar URL: http://res-dom.iana.org
   Updated Date: 2026-01-16T18:26:50Z
   Creation Date: 1995-08-14T04:00:00Z
   Registry Expiry Date: 2026-08-13T04:00:00Z
   Registrar: RESERVED-Internet Assigned Numbers Authority
   Registrar IANA ID: 376
   Registrar Abuse Contact Email:
   Registrar Abuse Contact Phone:
   Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
   Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
   Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
   Name Server: ELLIOTT.NS.CLOUDFLARE.COM
   Name Server: HERA.NS.CLOUDFLARE.COM
   DNSSEC: signedDelegation
   DNSSEC DS Data: 2371 13 2 C988EC423E3880EB8DD8A46FE06CA230EE23F35B578D64E78B29C3E1C83D245A
   URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-02-02T14:59:27Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar.  Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
```

```
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this Data is expressly
prohibited without the prior written consent of VeriSign. You agree not to
use electronic processes that are automated and high-volume to access or
query the Whois database except as reasonably necessary to register
domain names or modify existing registrations. VeriSign reserves the right
to restrict your access to the Whois database in its sole discretion to ensure
operational stability.  VeriSign may restrict or terminate your access to the
Whois database for failure to abide by these terms of use. VeriSign
reserves the right to modify these terms at any time.

The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
% IANA WHOIS server
% for more information on IANA, visit http://www.iana.org
% This query returned 1 object

domain:       EXAMPLE.COM

organisation: Internet Assigned Numbers Authority

created:      1992-01-01
source:       IANA

arnav@arnav-HP-Laptop-15s-fq5xxx:~$
```

17. ifplugstatus

Description: ifplugstatus is a lightweight utility used to detect the link status of a local network interface. It tells you whether a physical Ethernet cable is actually plugged into the network card and if the hardware has detected a carrier signal (link beat).

Output:

```
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ ifplugstatus
lo: link beat detected
wlo1: link beat detected
arnav@arnav-HP-Laptop-15s-fq5xxx:~$
```

18. nload

Description: nload is a console application which monitors network traffic and bandwidth usage in real time. It visualizes the ingoing and outgoing traffic using two

graphs and provides additional information like the total amount of transferred data and min/max network usage.

Output:

```
Device wlo1 [192.168.1.14] (2/2):
==========================================================================================
Incoming:
                                                                    Curr: 72.66 kBit/s
                                                                    Avg: 9.22 kBit/s
                                                                    Min: 0.00 Bit/s
                                                                    Max: 181.30 kBit/s
                                                                    Ttl: 624.10 MByte
Outgoing:
                                                                    Curr: 63.92 kBit/s
                                                                    Avg: 16.55 kBit/s
                                                                    Min: 0.00 Bit/s
                                                                    Max: 555.55 kBit/s
                                                                    Ttl: 45.61 MByte
```

19. mail

Description: mail is a command-line utility used to send, receive and manage emails directly from the terminal.

Output:

```
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ mail arnavsinghal2607@gmail.com
Cc:
Subject: Test Mail
Sample text
```

20. hostname

Description: hostname is used to display the system's DNS name, and to display or set its hostname or NIS domain name.

Output:

```
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ hostname
arnav-HP-Laptop-15s-fq5xxx
arnav@arnav-HP-Laptop-15s-fq5xxx:~$
```

21. mtr

Description: mtr is a powerful network diagnostic tool that combines the functions of ping and traceroute. It constantly probes the routers on the path between the local host and a destination, providing a real-time, scrolling view of the network performance and packet loss for every hop.

Output:
(Command: mtr google.com)

```
                                   My traceroute  [v0.95]
arnav-HP-Laptop-15s-fq5xxx (2401:4900:8fc8:a787:65c4:48e:d67f:b45d) -> google.com 2026-02-02T20:50:31+0530
Keys:  Help   Display mode   Restart statistics   Order of fields   quit
                                                          Packets               Pings
 Host                                                    Loss%   Snt   Last   Avg  Best  Wrst StDev
 1. 2401:4900:8fc8:a787:b6a7:c6ff:fe3d:6880               0.0%   114    3.8   5.2   1.9  61.9   7.5
 2. (waiting for reply)
 3. (waiting for reply)
 4. 2404:a800:2a00:101::9                                 0.0%   114    7.3  10.2   4.7 129.4  16.8
 5. 2404:a800::167                                        0.0%   114    8.2  11.6   7.1 132.4  15.8
 6. 2404:6800:8281::1                                     0.0%   113   10.5  11.2   7.5 109.6  11.1
 7. 2404:6800:8281::1                                     0.9%   113    8.9  12.9   7.5 246.1  24.5
 8. 2001:4860:0:1::7ba4                                   0.0%   113   11.4  14.4   9.2 148.6  18.3
 9. 2001:4860:0:1::7ba9                                   0.0%   113   10.1  12.2   8.6  98.7   9.3
10. bom07s01-in-x0e.1e100.net                             0.0%   113   10.1  12.4   9.2  72.1   9.0
```

22. nmap

Description: nmap is an open-source utility used for network discovery, inventory management and security auditing. It works by sending specially crafted packets to a target host and then analyzing the responses to determine what hosts are available on the network, what services (application name and version) those hosts are offering, what operating systems (and OS versions) they are running, what type of packet filters/firewalls are in use, and dozens of other characteristics.

Output:

```
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ nmap google.com
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-02-02 20:52 IST
Nmap scan report for google.com (142.250.71.110)
Host is up (0.0099s latency).
Other addresses for google.com (not scanned): 2404:6800:4009:806::200e
rDNS record for 142.250.71.110: pnbomb-ad-in-f14.1e100.net
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 4.05 seconds
arnav@arnav-HP-Laptop-15s-fq5xxx:~$
```

23. ssh

Description: ssh is a cryptographic network protocol and utility used for operating network services securely over an unsecured network. Its most common use is for remote login to a computer system, providing an encrypted channel for executing commands and moving data between a local and remote host.

Output:

```
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ ssh arnav@192.168.1.17
arnav@192.168.1.17's password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-139-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Infrastructure is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

169 additional security updates can be applied with ESM Infra.
Learn more about enabling ESM Infra service for Ubuntu 20.04 at
https://ubuntu.com/20-04

New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Your Hardware Enablement Stack (HWE) is supported until April 2025.
Last login: Mon Feb  2 23:10:35 2026 from 192.168.1.14
arnav@arnav-desktop-1:~$ hostname -I
192.168.1.17 2401:4900:8fc8:a787:5816:3c44:26d3:b1ff 2401:4900:8fc8:a787:123d:a9cc:e706:2941
arnav@arnav-desktop-1:~$ logout
Connection to 192.168.1.17 closed.
```

24. nmcli

Description: nmcli is a command-line tool used for controlling NetworkManager and reporting network status. It is used to create, display, edit, delete, activate and deactivate network connections, as well as control and display network device status.

Output:

```
arnav@arnav-HP-Laptop-15s-fq5xxx:~$ nmcli
wlo1: connected to Auto Airtel_Technologicle
        "Realtek RTL8822CE"
        wifi (rtw_8822ce), A8:41:F4:AF:AD:31, hw, mtu 1500
        ip4 default, ip6 default
        inet4 192.168.1.14/24
        route4 192.168.1.0/24 metric 600
        route4 default via 192.168.1.1 metric 600
        inet6 2401:4900:8fc8:a787:65c4:48e:d67f:b45d/64
        inet6 2401:4900:8fc8:a787:740b:8ff:c524:75c3/64
        inet6 fe80::b7cd:6fb0:ac76:b988/64
        route6 fe80::/64 metric 1024
        route6 2401:4900:8fc8:a787::/64 metric 600
        route6 default via fe80::1 metric 600

lo: connected (externally) to lo
        "lo"
        loopback (unknown), 00:00:00:00:00:00, sw, mtu 65536
        inet4 127.0.0.1/8
        inet6 ::1/128

E4:EC:E8:AB:7B:E1: disconnected
        "Arnav's M14"
        1 connection available
        bt (bluez), E4:EC:E8:AB:7B:E1, hw

p2p-dev-wlo1: disconnected
        "p2p-dev-wlo1"
        wifi-p2p, hw
```

```
DNS configuration:
        servers: 192.168.1.1
        interface: wlo1

Use "nmcli device show" to get complete information about known devices and
"nmcli connection show" to get an overview on active connection profiles.

Consult nmcli(1) and nmcli-examples(7) manual pages for complete usage details.
arnav@arnav-HP-Laptop-15s-fq5xxx:~$
```